

# SOLUTION:

## 1. c) 2946

$$40 \times 8 + 6 = 326$$

$$326 \times 9 + 7 = 294$$

$$2946 \times 10 + 8 = 29418$$

$$29418 \times 11 + 9 = 323607$$

## 2. b) 1725

the difference between numbers is  $+23^2$

$$+25^2, +27^2, +29^2, \dots$$

## 3. d) 3252

the difference between numbers is  $-(13^2+1)$ ,

$$-(11^2+1), -(9^2+1), -(7^2+1), -(5^2+1)$$

## 4. c) 90

$$2442/2 + 1 = 1222$$

$$1222/2 + 3 = 614$$

$$614/2 + 5 = 312$$

$$312/2 + 7 = 163$$

$$163/2 + 9 = 89.5$$

$$89.5/2 + 11 = 55.75$$

## 5. e) 1250

the difference between numbers is  $+(3^3+3)$

$$+(4^3+4), +(5^3+5), +(6^3+6), +(7^3+7),$$

$$+(8^3+8)$$

## 6. c) 50

$$1953.125/2.5 = 781.25$$

$$781.25/2.5 = 312.5$$

$$312.5/2.5 = 125$$

$$125/2.5 = 50$$

$$50/2.5 = 20$$

## 7. d) 59904

the difference between numbers is  $*1+6, *2-$

$$12, *3+18, *4-24, *5+30, *6-36$$

## 8. c) 210

the difference between numbers is  $+23, +46,$

$$+92, +184, +368, +736$$

## 9. b) 658

the difference between numbers is  $+1^3+0,$

$$+2^3+5, +3^3+10, +4^3+15, +5^3+20$$

$$+6^3+25$$

## 10. c) 928

the difference between numbers is

$$+2^3-2^2$$

$$+3^3-3^2$$

$$+4^3-4^2$$

$$+5^3-5^2$$

$$+6^3-6^2$$

$$+7^3-7^2$$

## 11. c) 309

$$7 \times 1 - 2 = 5$$

$$5 \times 2 - 3 = 7$$

$$7 \times 3 - 4 = 17$$

$$17 \times 4 - 5 = 63$$

$$63 \times 5 - 6 = 309$$

## 12. a) 52

$$50..50+(1^3+1) = 52$$

$$52+(2^3+1)=61$$

$$61+(3^3+1)=89$$

$$89+(4^3+1)=154$$

$$154 + (5^3+1) = 280$$

## 13. c) 57

$$17 + 1 \times 2 = 19$$

$$19 + 2 \times 3 = 25$$

$$25 + 3 \times 4 = 37$$

$$37 + 4 \times 5 = 57$$

$$57 + 5 \times 6 = 87$$

## 14. e) 66

$$3....5....9....15....23$$

$$...2....4....6....8....$$

$$\text{Answer } 43+23 = 66$$

## 15. b) 1182

$$26 \times 6 - 12 = 144$$

$$144 \times 4 + 14 = 590$$

$$590 \times 2 - 16 = 1164$$

$$1164 \times 1 + 18 = 1182$$



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NUMBER SERIES



$$4 \times 1 = 4$$

$$4 \times 1.5 = 6$$

$$6 \times 2 = 12$$

$$12 \times 2.5 = 30$$

$$30 \times 3 = 90$$

$$90 \times 3.5 = 315$$

43. c) 136

$$3 \times 1 + 1 = 4$$

$$4 \times 2 + 2 = 10$$

$$10 \times 3 + 3 = 33$$

$$33 \times 4 + 4 = 136$$

$$135 \times 5 + 5 = 685$$

$$685 \times 6 + 6 = 4116$$

44. e) 285

$$2 \times 1 + 1 = 3$$

$$3 \times 2 - 2 = 4$$

$$4 \times 3 + 3 = 15$$

$$15 \times 4 - 4 = 56$$

$$56 \times 5 + 5 = 285$$

$$285 \times 6 - 6 = 1704$$

45. d) 205.5

$$6 \times 1 + 1 = 7$$

$$7 \times 1.5 + 1.5 = 12$$

$$12 \times 2 + 2 = 26$$

$$26 \times 2.5 + 2.5 = 67.5$$

$$67.5 \times 3 + 3 = 205.5$$

46. e) 1610

$$*1 + 2, *2 + 4, *3 + 6, *4 + 8, *5 + 10, *6 + 12, \dots$$

47. b) 442

$$1 + 0 \times 3 = 1$$

$$1 + 1 \times 3 = 2$$

$$2 + 2 \times 3 = 10$$

$$10 + 3 \times 3 = 37$$

$$37 + 4 \times 3 = 101$$

$$101 + 5 \times 3 = 226$$

$$226 + 6 \times 3 = 442$$

48. c) 136

$$3 \times 1 + 1 = 4$$

$$4 \times 2 + 2 = 10$$

$$10 \times 3 + 3 = 33$$

$$33 \times 4 + 4 = 136$$

$$136 \times 5 + 5 = 685$$

$$685 \times 6 + 6 = 4116$$

49. d) 536

$$4 \times 1 + 1 = 5$$

$$5 \times 2 - 2 = 8$$

$$8 \times 3 + 4 = 28$$

$$28 \times 4 - 8 = 104$$

$$104 \times 5 + 16 = 536$$

50. a) 135

$$12, 15, 25, 42, 66, 97, ?$$

$$\begin{array}{ccccccc} \vee & \vee & \vee & \vee & \vee & \vee & \\ 3 & 10 & 17 & 24 & 31 & 38 & \\ & \vee & \vee & \vee & \vee & \vee & \\ & 7 & 7 & 7 & 7 & 7 & \end{array}$$

There should be  $97 + 38 = 135$

51. d) 872

$$*1 + 2, *2 + 4, *3 + 8, *4 + 16, *5 + 32$$

52. a) 35

$$7 + 4 = 11$$

$$11 + 8 = 19$$

$$19 + 16 = 35$$

$$35 + 32 = 67$$

$$67 + 64 = 131$$

53. c) 10

$$17 \times 0.5 + 0.5 = 9$$

$$9 \times 1 + 1 = 10$$

$$10 \times 1.5 + 1.5 = 16.5$$

$$16.5 \times 2 + 2 = 35$$

$$35 \times 2.5 + 2.5 = 90$$

54. c) 45

$$5 \times 0.5 + 0.5 = 3$$

$$3 \times 1 + 1 = 4$$

$$4 \times 1.5 + 1.5 = 7.5$$

$$7.5 \times 2 + 2 = 17$$

$$17 \times 2.5 + 2.5 = 45$$

55. a) 133

$$9 + 11 = 20$$

$$11 + 20 = 31$$

$$31 + 51 = 82$$

$$82 + 51 = 133$$

56. b) 96

$$5 + (1^2) = 5 + 1 = 6$$

$$6 + (2^2) = 6 + 4 = 10$$

$$10 + (3^2) = 10 + 9 = 19$$

$$19 + (4^2) = 19 + 16 = 35$$

$$35 + (5^2) = 35 + 25 = 60$$

$$60 + (6^2) = 60 + 36 = 96$$

57. c) 46

$$24 + 2^2 = 24 + 4 = 28$$

$$28 - 3^2 = 28 - 9 = 19$$

$$19 + 4^2 = 19 + 16 = 35$$

$$35 - 5^2 = 35 - 25 = 10$$

$$10 + 6^2 = 10 + 36 = 46$$

58. b) 75

2, 5, 9, 19, 37, ..... The pattern is: every number is arrived at previous number multiplied by 2 and then alternate addition and subtraction by 1 i.e.

$$2$$

$$5 = 2 \times 2 + 1$$

$$9 = 5 \times 2 - 1$$

$$19 = 9 \times 2 + 1$$

$$37 = 19 \times 2 - 1$$

$$\text{the next term } 37 \times 2 + 1 = 75$$

59. b) 35

$$9 = 4 \times 2 + 1$$

$$17 = 9 \times 2 - 1$$

$$35 = 17 \times 2 + 1$$

$$69 = 35 \times 2 - 1$$

$$139 = 69 \times 2 + 1$$

$$277 = 139 \times 2 - 1$$

60. d) 134

$$3 \times 7 + 1 = 22$$

$$22 \times 6 + 2 = 134$$

$$134 \times 5 + 3 = 532$$

$$532 \times 4 + 4 = 2132$$

61. d) 610

$$5 \times 3 - 5 = 10$$

$$10 \times 3 - 5 = 25$$

$$25 \times 3 - 5 = 70$$

$$70 \times 3 - 5 = 205$$

$$205 \times 3 - 5 = 610$$

62. c) 1050

$$7 \times 5 = 35$$

$$35 \times 6 = 210$$

$$210 \times 5 = 1050$$

$$1050 \times 6 = 6300$$

$$6300 \times 5 = 31500$$

63. a) 129

$$6 \times 1 + 1 \times 7 = 13$$

$$13 \times 2 + 2 \times 6 = 38$$

$$38 \times 3 + 3 \times 5 = 129$$

$$129 \times 4 + 4 \times 4 = 532$$

$$532 \times 5 + 5 \times 3 = 2675$$

64. b) 109

$$7, 14, 25, 40, 59, 82, 109$$

$$\begin{array}{ccccccc} \vee & \vee & \vee & \vee & \vee & \vee & \\ 7 & 11 & 15 & 19 & 23 & 27 & \\ & \vee & \vee & \vee & \vee & \vee & \\ & 4 & 4 & 4 & 4 & 4 & \end{array}$$

There should be  $82 + 27 = 109$

65. a) 224

$$5 + 1^3 = 6$$

$$6 + 2^2 = 10$$

$$10 + 3^3 = 37$$

$$37 + 4^2 = 53$$

$$53 + 5^3 = 188$$

$$188 + 6^2 = 224$$

66. d) 630

$$2 \times 1 = 2$$

$$2 \times 1.5 = 3$$

$$3 \times 2 = 6$$

$$6 \times 2.5 = 15$$

$$15 \times 3 = 45$$

$$45 \times 3.5 = 157.5$$

$$157.5 \times 4 = 630$$

67. d) 183

$$9 \times 5 + 5 = 50$$

$$5 \times 1 + 1 = 6$$

$$6 \times 1.5 + 1.5 = 10.5$$

$$10.5 \times 2 + 2 = 23$$

$$23 \times 2.5 + 2.5 = 60$$

$$60 \times 3 + 3 = 183$$

68. a) 155



16, 20, 29, 45, 70, 106, ?  
 $\begin{array}{ccccccc} \vee & \vee & \vee & \vee & \vee & \vee \\ 4 & 9 & 16 & 25 & 36 & 49 \end{array}$

Squares of consecutive numbers, the next term should be  $106+49 = 155$

69. c) 1870

$7*1+5 = 12$   
 $12*2+5 = 29$   
 $29*3+5 = 92$   
 $92*4+5 = 373$   
 $373*5+5 = 1870$

70. a) 124

4, 9, 20, 37, 60, 89, ?  
 $\begin{array}{ccccccc} \vee & \vee & \vee & \vee & \vee & \vee \\ 5 & 11 & 17 & 23 & 29 & 35 \\ \vee & \vee & \vee & \vee & \vee & \vee \\ 6 & 6 & 6 & 6 & 6 & 6 \end{array}$

There should be  $89+35 = 124$

71. a) 32.5

$4*1+1 = 5$   
 $5*1.5-1.5 = 6$   
 $6*2+2 = 14$   
 $14*2.5-2.5 = 32.5$   
 $32.5*3+3 = 100.5$

72. e) 256

$8*0.5 = 4$   
 $4*1 = 4$   
 $4*2 = 8$   
 $8*4 = 32$   
 $32*8 = 256$

73. e) 22

$2+1^2-1 = 2$   
 $2+2^2+1 = 7$   
 $7+4^2-1 = 22$

74. d) 47

$6*1+2 = 8$   
 $8*1.5-4 = 8$   
 $8*2+6 = 22$   
 $22*2.5-8 = 47$   
 $47*3+10 = 151$

75. c) 6

$4+1^2+1 = 6$   
 $6+3^2-1 = 14$   
 $14+5^2+1 = 40...$

76. c) 25

$6*1-1+1 = 6$   
 $6*2-2-3 = 7$   
 $7*3-1+5 = 25$

77. c) 257

$2+1^3+2 = 5$   
 $5+2^3+4 = 17$   
 $17+3^3+6 = 50$   
 $50+4^3+8 = 122$

78. d) 651

$2*4+1 = 9$   
 $9*4+3 = 39$   
 $39*4+5 = 161$   
 $161*4+7 = 651$   
 $651*4+9 = 2613$

79. d) 8

$5*2-2 = 8$   
 $8*2+4 = 20$   
 $20*2-6 = 34$

80. a) 645

$5*1+1 = 6$   
 $6*2-2 = 10$   
 $10*3+3 = 33$   
 $33*4-4 = 128$   
 $128*5+5 = 645$

81. a) 82.5

$8*1+1 = 9$   
 $9*1.5+1.5 = 15$   
 $15*2+2 = 32$   
 $32*2.5+2.5 = 82.5$   
 $82.5*3+3 = 250.5$

82. b) 784

$12*0.5+1 = 7$   
 $7*1+2 = 9$   
 $9*2+4 = 22$   
 $22*4+8 = 96$   
 $96*8+16 = 784$

83. b) 24

$2+1^2+1 = 4$   
 $4+2^2-1 = 7$   
 $7+4^2+1 = 24$   
 $24+8^2-1 = 87$   
 $87+16^2+1 = 344$

84. a) 119

$6*1-2 = 4$   
 $4*1.5+4 = 10$   
 $10*2-6 = 14$   
 $14*2.5+8 = 43$   
 $43*3-10 = 119$

85. c) 6

$4+1^2+1 = 6$   
 $6+3^2-1 = 14$   
 $14+5^2+1 = 40$   
 $40+7^2-1 = 88$   
 $88+9^2+1 = 170$

86. d) 25

$6*1-1+1 = 6$   
 $6*2-2-3 = 7$   
 $7*3-1+5 = 25$   
 $25*4-2-7 = 91$   
 $91*5-1+9 = 463$

87. a) 233

$2+1^3+2 = 5$   
 $5+2^3-4 = 9$   
 $9+3^3+6 = 42$   
 $42+4^3-8 = 98$   
 $98+5^3+10 = 233$

88. e) 739

$3*4-1 = 11$   
 $11*4+3 = 47$   
 $47*4-5 = 183$   
 $183*4+7 = 739$   
 $739*4-9 = 2947$

89. a) 4

$2*2-2 = 2$   
 $2*2+4 = 8$   
 $8*2-6 = 10$   
 $10*2+8 = 28$   
 $28*2-10 = 46$

90. e) 570

$4*1+2 = 6$   
 $6*2-4 = 8$   
 $8*3+6 = 30$   
 $30*4-8 = 112$   
 $112*5+10 = 570$

91. d) 94.75

$(19*0.5)+1 = 10.5$   
 $(10.5*1)+1 = 11.5$   
 $(11.5*1.5)+1 = 18.25...$   
 Correct Number: 94.75

92. c) 22

$21+1^2+1 = 23$   
 $23+2^2+1 = 28$   
 $28+3^2+1 = 38 .....$   
 Correct Number: 23

93. b) 32

$3*1+1 = 4$   
 $4*2+2 = 10$   
 $10*3+3 = 33.....$   
 Correct Number: 33

94. d) 10

$4*3 = 12$   
 $12*4 = 48$   
 $48*5 = 240.....$   
 Correct Number: 12

95. e) None of these

First series: 3, 10, 17, 24 (increased by 7)  
 Second Series: 5, 14, 23 (increased by 9)  
 Wrong Number: 12  
 Correct Number: 14

96. d) 38

First series: 3, 6, 9 (increased by 3)  
 Second Series: 7, 28, 112, 448 (multiplied by 4)  
 Wrong Number: 38  
 Correct Number: 9

97. a) 28

First series: 650, 130, 32.5, 10.8 (divided by 5, 4, 3)  
 Second Series: 148, 108, 68 (difference : 40)



Wrong Number: 28

Correct Number: 68

**98. b) 520**

$7^4 - 6, 6^4 - 6, 5^4 - 5, 4^4 - 4, \dots$

Correct Number: 620

**99. d) 1538**

$19 \times 1 - (1 \times 7) = 12$

$12 \times 2 - (2 \times 6) = 12$

$12 \times 3 - (3 \times 5) = 21$

$21 \times 4 - (4 \times 4) = 68, \dots$

Correct Number: 1838

**100. c) 45**

$2580 \times 0.25 = 645$

$645 \times 0.5 = 322.5$

$322.5 \times 0.25 = 80.62$

$80.62 \times 0.5 = 40.31, \dots$

Correct Number: 40.31

**101. c) 325**

$9^2 - 1 = 80 \Rightarrow 80 + 105 = 185$

$8^2 - 2 = 62 \Rightarrow 185 + 62 = 247$

$6^2 - 4 = 32 \Rightarrow 293 + 32 = 325$

**102. b) 440**

Explanation :

$5 \times 1 = 5 + 5 = 10$

$10 \times 2 = 20 + 10 = 30$

$105 \times 4 = 420 + 20 = 440$

**103. a) 15**

$4 \times 0.5 = 2$

$2 \times 1 = 2$

$2 \times 1.5 = 3$

$6 \times 2.5 = 15$

**104. d) 347**

$5 \times 3 - 1 = 14$

$14 \times 3 - 2 = 40$

$117 \times 3 - 4 = 347$

**105. c) 8800**

$15 \times 2 - 8 = 22$

$22 \times 4 - 16 = 72$

$72 \times 8 - 24 = 552$

$552 \times 16 - 32 = 8800$

**106. a) 93.5**

$12 \times 1.5 + 2 = 20$

$20 \times 1.5 + 4 = 34$

$34 \times 1.5 + 6 = 57$

$57 \times 1.5 + 8 = 93.5$

**107. d) 2.5**

$20 - 4 \div 2 = 8$

$8 + 4 \div 2 = 6$

$6 - 4 \div 2 = 1$

$1 + 4 \div 2 = 2.5$

**108. b) 970**

$1200 - 81 = 1119$

$1119 - 64 = 1055$

$1055 - 49 = 1006$

$1006 - 36 = 970$

**109. d) 2276**

$2561 - 121 = 2440$

$2440 - 81 = 2359$

$2359 - 49 = 2310$

$2310 - 25 = 2285$

$2285 - 9 = 2276$

**110. c) 1576**

$1 \times 7 + 7 = 14$

$14 \times 6 - 6 = 78$

$78 \times 5 + 5 = 395$

$395 \times 4 - 4 = 1576$